FIELDS GENERATED BY A CLASSIFYING INVARIANT OF CM BIELLIPTIC PICARD CURVES

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Abstract. We calculate the fields generated by certain curves of the kind $C\colon y^3 = x^4 + ax^2 + b$. The associated Prym variety P of this kind of curves admits an embedding of a quaternion order $\mathcal{O}$, and we are interested in the cases where the centralizer of $\mathcal{O}$ in $\mathrm{End}(P)$ is a maximal order of an imaginary quadratic field. We characterize the imaginary quadratic fields, K, whose maximal order is the centralizer of $\mathcal{O}$ in $\mathrm{End}(P)$ for some C of the above kind, and we calculated the field generated over K by a classifying invariant of the curves of the above kind in terms of class field theory.

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1. introduction

A **bielliptic Picard curve** $C/\mathbb{Q}$, is a smooth projective curve over $\mathbb{Q}$ with an affine equation $y^3 = x^4 + ax^2 + b$. Laga and Shnidman defined in [4, §2.3] a j-invariant for each bielliptic picard curve $C/\mathbb{Q}$ by

$$j(C) := \frac{4b - a^2}{4b}$$

and showed that given another bielliptic picard curve $C'/\mathbb{Q}$, $C' \simeq C$ if and only if $j(C') = j(C)$. This paper is concerned with the values of this invariant for a special class of bielliptic picard curves. To introduce this class, note that $(x, y) \mapsto (x^2, y)$ defines a map, π, from C to $E : y^3 = x^2 + ax + b$. The map π induces a morphism of Abelian varieties $\pi_* : J(C) \to J(E) \simeq E$ where $J(C), J(E)$ are the Jacobian varieties of C, E respectively. We define the **Prym** of C, P, to be the kernel of π_*. Laga and Shnidman showed in [4, Corollary 4.8] that there is an embedding ι of the maximal order $\mathcal{O}$ of the quaternion algebra $\left(\frac{-3,2}{\mathbb{Q}}\right)$, in $\mathrm{End}(P)$. In light of this, we define

$$\mathrm{End}_{\mathcal{O}}(P) := \{f \in \mathrm{End}(P) | f \circ \iota(b) = \iota(b) \circ f \ \forall b \in \mathcal{O}\}$$

and we will show in lemma 2.5 that it is independent of the choice of ι. According to [4, §6.4], $\mathrm{End}_{\mathcal{O}}(P)$ is either $\mathbb{Z}$ or an order in an imaginary quadratic field. In the latter case, we say that C is CM or has **CM** by $\mathrm{End}_{\mathcal{O}}(P)$. The bielliptic picard curves with CM by maximal quadratic orders are the ones that we will be interested in the values of their j-invariants.

The main aim of the paper is to establish a relationship between the field generated over K by j invariant of a bielliptic Picard curve with CM by $\mathcal{O}_K$, when $K/\mathbb{Q}$ is an imaginary quadratic extension and $\mathcal{O}_K$ is its maximal order, to the Hilbert class field of $\mathcal{O}_K$, H. However, to state this relationship, we first need to characterize the set of maximal quadratic orders by which some bielliptic picard curve has CM. Let $K/\mathbb{Q}$ be an imaginary quadratic field, the first result of the paper is the following

Theorem 1.1. *There exists a bielliptic Picard curve C with CM by $\mathcal{O}_K$ if and only if K does not split at 2 and does not split at 3 and $K \neq \mathbb{Q}[\sqrt{-3}]$.*

See Section 3 for the proof. Let $C/\mathbb{Q}$ be a bielliptic Picarc curve with CM by $\mathcal{O}_K$. We would like to phrase the relationship that we wish to establish, in terms of the Galois group of the field generated over K by the j-invariant of C. To do it in a convenient way, we would like to first find a field that always contains $K[j(C)]$ and is easy to describe, and this constitutes the second result of the paper:

Theorem 1.2. $K[j(C)] \subseteq H[\sqrt{-3}]$

see section 4 for the proof. In light of this result, we can describe the field generated by $j(C)$ over K by describing the following group:

$$G = \mathrm{Gal}(H[\sqrt{-3}]/K[j(C)])$$

This group in turn may be described by describing its image under the natural injection

$$\phi : \mathrm{Gal}(H[\sqrt{-3}]/K) \to \mathrm{Gal}(K[\sqrt{-3}]/K) \times \mathrm{Gal}(H/K)$$

and indeed the main result is:

Theorem 1.3. *Identify* $\mathrm{Gal}(K[\sqrt{-3}]/K)$ *with* $\{\pm 1\}$ *, let D be the discriminant of K and denote*

$$\sigma = \begin{cases} \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right) & 2|D \\ 1 & 2 \not| D \end{cases}$$

$$\tau = \begin{cases} \left(1, \left(\frac{H/K}{\mathfrak{a}}\right)\right) & D \equiv 6 \bmod 9 \\ 1 & D \not\equiv 6 \bmod 9 \end{cases}$$

where $\mathfrak{a}, \mathfrak{b}$ are the unique primes lying over $3, 2$ respectively, and $\left(\frac{H/K}{\mathfrak{b}}\right), \left(\frac{H/K}{\mathfrak{a}}\right)$ are the Artin symbols of $\mathfrak{b}, \mathfrak{a}$ respectively. Then

$$\phi\left(G\right) = \langle \sigma, \tau \rangle$$

To unpack everything, we have shown that $K[j(C)]$ is the field that the Galois group of $H[\sqrt{-3}]$ over it is generated by the preimages under ϕ of σ and τ. See section 5 for the proof.

Finally in section 6 we present some examples of low degree j-invariants and the fields they generate and in section 7 we suggest a few directions for further research.

2. Background

2.1. **Splitting of Primes in Quadratic Extensions of $\mathbb{Q}$.** Let K be an imaginary quadratic field. We recall the usual definition:

Definition 2.1. *A prime $p \in \mathbb{Z}$ splits in K if there is more than one prime in $\mathcal{O}_K$ lying over p.*

This definition is the one that lets us define the primes $\mathfrak{b}, \mathfrak{a}$ in the main result, 5.10, as the unique primes lying over $2, 3$ respectively. We will show that it is equivalent to another definition which will be useful in Section 3:

Lemma 2.2. *For every prime p in $\mathbb{Z}$, $K_p := K \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is not a field if and only if p splits in K.*

Proof. Let n be a square free integer such that $K = \mathbb{Q}[\sqrt{-n}]$. First note that

$$K_p \simeq \frac{\mathbb{Q}_p[x]}{(x^2+n)}$$

Indeed, since $\mathbb{Q}_p$ and K are both commutative $\mathbb{Q}$-algebras their tensor product is their co-product in the category of commutative $\mathbb{Q}$-algebras. However, as functors from the category of commutative $\mathbb{Q}$-algebras to Set

$$\begin{aligned}
&X \mapsto \hom\left(\frac{\mathbb{Q}_p[x]}{(x^2+n)}, X\right) \simeq \\
&X \mapsto \left\{f \in \hom\left(\mathbb{Q}_p[x], X\right) : (x^2+n) \subseteq \ker(f)\right\} \simeq \\
&X \mapsto \left\{(f,x) \in \hom\left(\mathbb{Q}_p, X\right) \times X : x^2+n = 0\right\} \simeq \\
&X \mapsto \hom\left(\mathbb{Q}_p, X\right) \times \left\{x \in X : x^2+n = 0\right\} \simeq \\
&X \mapsto \hom\left(\mathbb{Q}_p, X\right) \times \left\{f \in \hom\left(\mathbb{Q}[x], X\right) : (x^2+n) \subseteq \ker(f)\right\} \simeq \\
&X \mapsto \hom\left(\mathbb{Q}_p, X\right) \times \hom\left(K, X\right)
\end{aligned}$$

and hence by the full faithfulness of the Yoneda embedding, $\frac{\mathbb{Q}_p[x]}{(x^2+n)}$ is isomorphic to the co-product of $\mathbb{Q}_p$ and X in the category of commutative Q-algebras.

Therefore, it's enough to show that p splits in K if and only if $\frac{\mathbb{Q}_p[x]}{(x^2+n)}$ isn't a field i.e. if and only if x^2+n is reducible in $\mathbb{Q}_p[x]$ or equivalently $-n \in {\mathbb{Q}_p}^2$. We will divide into two cases:

- $2 \nmid p$. If p splits in K then x^2+n splits over $\mathbb{Z}/p$ and hence by Hensel's lemma it is reducible over $\mathbb{Q}_p$ as required. Assume that $-n \in {\mathbb{Q}_p}^2$. Then $2|v_p(-n)$ and hence, since $-n$ is square free, $p \nmid n$. Therefore $-n$ reduces to an invertible square in $\mathbb{Z}/p$ and hence p is split in K as required.
- $2 = p$. Assume that 2 splits in K. Therefore $-n \equiv 1 \bmod 8$ and hence there is $a \in \mathbb{Z} \subseteq \mathbb{Z}_2$ such that $a^2+n \equiv 0 \bmod 8$. In particular it follows that $2 \nmid a$, and hence

  $$2v_2\left(\frac{\partial(x^2+n)}{\partial x}|_{x=a}\right) = 2v_2(2a) = 2 \cdot 1 = 2 < 3 \leq v_2(a^2+n)$$

  and hence, by a stronger version of Hensel's lemma, x^2+n has a root in $\mathbb{Z}_2$ or equivalently is reducible in $\mathbb{Z}_p[x]$ as required.

  Assume that $-n \in {\mathbb{Q}_2}^2$. By the same argument as in the general case above, $2 \nmid n$ and hence $-n$ reduces to an invertible square in $\mathbb{Z}/8$. However the only such element in $\mathbb{Z}/8$ is $1 + 8\mathbb{Z}$ and hence 2 splits in K as required.

□

Even though the above lemma is all we will need, we will mention without proof the well known generalization of it:

Proposition 2.3. *Let j be an algebraic extension of $\mathbb{Q}$ and p a prime in $\mathbb{Z}$. Then*

$$j \otimes_{\mathbb{Q}} \mathbb{Q}_p \simeq \prod_{\mathfrak{p}\in \mathrm{Spec}(O_L), \mathfrak{p}|pO_L} L_{\mathfrak{p}}$$

where $L_{\mathfrak{p}}$ is the completion of j with respect to the $\mathfrak{p}$-adic valuation.

2.2. **Quaternionic Algebras .** In this subsection we will assemble, from Voight's book, [6], some definitions and results about quaternion algebras which will be used in this paper. In [6, §2.2] Voight defines quaternion algebras and gives some contructions of such: Let F be a field of characteristic 0. A **quaternion algebra** over a field F, is an F-algebra that is generated, as a vector space, by $1, i, j, ij$ for some i, j in the algebra that satisfy

$$i^2 = a \quad j^2 = b \quad ji = -ij$$

for some $a, b \in F^\times$. Consider for any $a, b \in F^\times$ the algebra

$$\frac{K\langle i, j\rangle}{(i^2 - a, j^2 - a, ij + ji)} \tag{2.1}$$

where $K\langle i, j\rangle$ is the free (non-commutative) algebra generated over K by i, j. One can show that this is a quaternion algebra over F, and it is denoted by $\left(\frac{a,b}{F}\right)$. Furthermore, for any field $F \subseteq K$ and for any $a, b \in F^\times$,

$$\left(\frac{a, b}{F}\right) \otimes_F K \simeq \left(\frac{a, b}{K}\right)$$

and in particular this tensor product is also a quaternion algebra.

Let B be a quaternion algebra over F. In [6, Main Theorem 5.4.4], Voight shows, among other things, that B is not a division ring if and only if $B \simeq M_2(F)$. In light of this we say that B is **split** if $B \simeq M_2(F)$, and that a field $F \subseteq K$ is a **splitting field** for B if $B \otimes_F K$ is split.

In [6, Chapter 14], Voight explores the notion of ramification of quaternions. We will present it only in the case where $F = \mathbb{Q}$. For a place v at $\mathbb{Q}$, we say that B ramifies at, or is ramified at, v if $B \otimes_{\mathbb{Q}} \mathbb{Q}_v$ isn't a division ring. We denote by $\mathrm{Ram}\ B$ the set of places in $\mathbb{Q}$ at which B is ramified. Note that lemma 2.2 implies an analogy between this definition and the definition of ramification of quadratic extension. Voight draws some deeper analogy but we will not expand on it. The following is a version of [6, Proposition 14.6.7] which is specialized to the context that will be useful for us and includes only the conditions that will be useful for us in section 3:

Proposition 2.4. *Let $K/\mathbb{Q}$ be a quadratic extension. Then the following are equivalent:*

i K *is a splitting field for* B

ii There is an embedding $K \hookrightarrow B$ *of* $\mathbb{Q}$*-algebras.*

iii Every $v \in \mathrm{Ram}\ B$ *does not split in* K*.*

Finally, we will introduce a few notations that will be fixed in the sequel. Let B be the quaternion algebra $\left(\frac{-3,2}{\mathbb{Q}}\right)$ and let i, j be as in the construction 2.1. In particular $i^2 = -3,\ j^2 = 2,\ ij = -ji$. According to [2, §3.1], $\mathrm{Ram}B = \{2, 3\}$. Additionally let $\omega := \frac{1}{2}(-1+i)$ and note that $\omega^2+\omega+1 = 0$ so ω is a primitive third root of unity. Using ω define

$$\mathcal{O} := \mathbb{Z}[\omega, j] = \mathbb{Z} + \mathbb{Z}\omega + \mathbb{Z}j + \mathbb{Z}\omega j$$

According to [4, §4.1] $\mathcal{O}$ is a maximal order, and since by [6, Corollary 10.5.5] all the maximal orders in B are conjugate, we will often refer to it as the maximal order of B as we have alrady done in the introduction.

2.3. **CM orders of bielliptic Picard curves.** A **quaternionic multiplication abelian surface** is a pair (A, ι) where A is an abelian surface and $\iota : \mathcal{O} \hookrightarrow \mathrm{End}(A)$ is a ring homomorphism. As in the introduction, we may define

$$\mathrm{End}_{\mathcal{O}}(A) := \{f \in \mathrm{End}(A) | f \circ \iota(b) = \iota(b) \circ f \ \forall b \in \mathcal{O}\}$$

Let C be a CM bielliptic Picard curve and let P be its prime. As noted in the introduction, Laga and Shnidamn proved in [4, Corollary 4.8] that there is an embedding $\iota : \mathcal{O} \hookrightarrow \mathrm{End}_{\mathcal{O}}(P)$. In [4, Lemma 6.2] they claim that P is not-simple, i.e. P is the product of two elliptic curves, if and only if $\mathrm{End}(P) \not\simeq \mathcal{O}$. In this case P and C is called CM because, according to [4, §6.4], $\mathrm{End}_{\mathcal{O}}(P)$ is an order in an imaginary quadratic field. As promised in the introduction we will show that this quadratic field is independent of the choice of ι. Let $K := \mathrm{End}_{\mathcal{O}}(P) \otimes_{\mathbb{Z}} \mathbb{Q}$.

Lemma 2.5. $\mathrm{End}_{\mathcal{O}}(P)$ *is independent of the choice of* ι.

Proof. According to [5, Theorem 2.10], there exist K-CM elliptic curves E_1, E_2 such that $P \simeq E_1 \times E_2$. Therefore,

$$\mathrm{End}(P) \otimes_{\mathbb{Z}} \mathbb{Q} \simeq M_2(K)$$

so we shall identify between the two. Let $\iota : \mathcal{O} \hookrightarrow \mathrm{End}(P)$. We can extend it to an embeddings

$$\iota : B \simeq \mathcal{O} \otimes \mathbb{Q} \hookrightarrow \mathrm{End}(P) \otimes_{\mathbb{Z}} \mathbb{Q} \simeq M_2(K)$$

The elements of $K \cdot I_2 \cap \mathrm{End}(P)$ commute with all of $M_2(K)$ and in particular with $\iota(\mathcal{O})$ and hence $K \cdot I_2 \cap \mathrm{End}(P) \subset \mathrm{End}_{\mathcal{O}}(P)$. For a small enough order $O \subset K$, $O\mathrm{Id} \subset \mathrm{End}(P)$, and hence $K \cdot I_2 \cap \mathrm{End}(P)$ contains an order, and consequently it is itself an order in K. On the other hand we already know that $\mathrm{End}_{\mathcal{O}}(P)$ is an order in K, so it cannot contain any element out of $K \cdot I_2$ since this would make it at lest three dimensional. Therefore, $K \cdot I_2 \cap \mathrm{End}(P) = \mathrm{End}_{\mathcal{O}}(P)$ and this is independent of ι. □

2.4. **Marked bielliptic picard curves and an apology.** Recall from the introduction that we have defined a bielliptic Picard curve $C/\mathbb{Q}$ to be a smooth with an affine equation

$$y^3 = x^4 + ax^2 + b$$

For any such curve, there is a faithful μ_6 action, γ, on C given by

$$\zeta \cdot (x, y) = (\zeta^3 x, \zeta^2 y)$$

where $\zeta \in \mu_6(\overline{\mathbb{Q}})$. Since Laga and Shnidman considered bielliptic Picard curves in a broader context in [4], the specification of the μ_6 action adds an actual extra data. For this reason they define the term **marked bielliptic picard curve** which in our limited context would mean a pair of the form (C, γ) where C is some bielliptic Picard curve and γ is defined as above. For reasons that can not be fully justified in retrospective, we will occasionally use the term marked bielliptic Picard curve (C, γ), but since as noted in our context γ is well defined by C, we will interchangeably use the term bielliptic Picard curve C.

2.5. **Moduli spaces of QM and μ_3 abelian surfaces.** Due to time constraints and the complexity of this topic, this subsection is less through than the rest of the paper. Given $\nu \in \mathcal{O}$ such that $\nu^2 \in \mathbb{Q}_{<0}$, we define an $(\mathcal{O}, \nu)$**-QM abelian surface** to be any triple (A, λ, ι) such that (A, λ) is a polarized abelian surface, and $\iota : \mathcal{O} \hookrightarrow \mathrm{End}(\mathcal{O})$ is an embedding of rings such that the Rosati involution restricts to $b \mapsto \nu\bar{b}\nu^{-1}$. Let $\mathcal{X}$ be the coarse moduli space $(\mathcal{O}, i)$-QM abelian surfaces where we recall that $i^2 = -3$.

Let $\mu \in \mathcal{O}$ such that $\mu^2 = -6$. It is implicit in [4] that there is a bijection between the set of isomorphism classes of $(\mathcal{O}, i)$-QM abelian surfaces and the set of isomorphism classes of $(\mathcal{O}, \mu)$-QM abelian surfaces that maps each triple (A, λ, ι) to a triple (A, λ', ι), in particular this map does not change A or ι. It is furthermore implicit in [4] that this bijection induces an equivalence between the previous moduli problem and the moduli problem of $(\mathcal{O}, \mu)$-QM abelian surfaces, and since the above map does not change A or ι, each point on $\mathcal{X}$ that corresponds to CM surface according to one of the above two moduli interpretations, will correspond to a surface with the same CM order according to the other moduli interpretations. We will use these facts to carelessly apply to the first moduli problem results regarding the second moduli problem which is more common in the existing literature.

Using Laga and Shnidman modular interpretation of $\mathcal{X}$, we may define on it the action of the **Atkin-Lehner group** $W = \{1, w_2, w_3, w_6\} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ on X: if $b \in \mathcal{O}$ is any element whose (reduced) norm has absolute value $d \in \{2, 3, 6\}$, then $w_d([A, \lambda, \iota]) = [A, \lambda, \iota \circ [b]]$, where $[b] \in \mathrm{Aut}(\mathcal{O})$ denotes conjugation by b. Furthermore, we will make the notation

$$\mathcal{X}^* := \mathcal{X}/W$$

Note that since the action of the Atkin-Lehner group on $\mathcal{X}$ doesn't change the surface itself and hence doesn't change the CM order, each point on $\mathcal{X}^*$ either lies either under no CM point or only under CM points of the same CM order and hence in the latter case we will say that the point on $\mathcal{X}^*$ has CM by the same order as any of its preimages on $\mathcal{X}$.

Following [4], define a μ_3**-abelian surface** to be a triple (A, λ, α) such that (A, λ) is a $(1, 2)$-polarized abelian surface and $\alpha : \mu_3 \to \mathrm{Aut}(A)$ is a $\mathrm{Gal}_{\mathbb{Q}}$ equivariant group homomorphism such that for every primitive third root of unity, ω:

(1) $\alpha(\omega)^\vee \circ \lambda \circ \alpha(\omega) = \lambda$

(2) $\alpha(\omega)$ has characteristic polynomial $X^2 + X + 1$ on $T_0 A = H^0(A, \Omega^1_A)^\vee$

Let $\mathcal{Y}$ be the coarse moduli space of μ_3 abelian surfaces. Finally we will call a μ_3 abelian surface **decomposable**, if it is the product of two elliptic curves.

As we will see in the next section, μ_3-abelian surface and bielliptic Picard curves are closely related. To set the ground for this relation, to each marked bielliptic picard curve (C, γ) we will associate a μ_3-abelian surface in the following way: As noted in the introduction, $(x, y) \mapsto (x^2, y)$ defines a map, π, from C to $E : y^3 = x^2 + ax + b$. The map π induces a morphism of Abelian varieties $\pi_* : J(C) \to J(E) \simeq E$, and we define the Prym of C, P, to be the kernel of π_*. The principal polarization on $J(C)$ restricts to a $(1, 2)$-polarization $\lambda : P \to P^\vee$. There is a μ_6 action on J that makes the Abel-Jacobi map $AJ_\infty : C \to J$ a μ_6 equivariant map (note that there is a unique point at ∞, namely $[0, 1, 0]$). This action induces a μ_3 action on P which we shall denote by α. To put it all together, we denote $\mathrm{Prym}\,(C, \gamma) = (P, \lambda, \alpha)$

3. The Maximal Quadratic Orders by Which Some Bielliptic Picard Curve has CM

Recall that K is an imaginary quadratic field.

Lemma 3.1. *If $B \hookrightarrow M_{2\times 2}(K)$ then $K \hookrightarrow B$*

Proof. Assume towards contradiction that $B \hookrightarrow M_{2\times 2}(K)$ but $K \not\hookrightarrow B$. According to Proposition 2.4, since $K \not\hookrightarrow B$, K doesn't split B, i.e. $B \otimes K \not\simeq M_{2\times_{\mathbb{Q}} 2}(K)$, i.e. $B \otimes_{\mathbb{Q}} K$ is a division ring. In particular, any homomorphism of rings (with unit) from $B \otimes_{\mathbb{Q}} K$ to another non-trivial ring is injective. Consequently, to derive a contradiction, it's enough to show that there is a homomorphism of $\mathbb{Q}$ algebras $B \otimes K \to M_{2\times 2}(K)$ since the dimension of both algebras over $\mathbb{Q}$ is 8 and hence it will follow that the two algebras are isomorphic and in particular $M_{2\times 2}(K)$ is a division ring and that's a contradiction. Due to the universal property of the tensor product of algebras, to derive the existance of such a map, it is enough to show that there are

$$f : B \to M_{2\times 2}(K)$$
$$g : K \to M_{2\times 2}(K)$$

such that $f(a)g(b) = g(b)f(a)$ for any $a \in B$ and $b \in K$. K embeds into the center of $M_{2\times 2}(K)$ so such a pair of maps indeed exists. □

Proposition 3.2. *There is a geometric $\mathcal{O}_K$-CM point on $\mathcal{X}$ if and only if K does not split at 2 nor at 3.*

Proof. Assume that there exists a geometric $\mathcal{O}_K$-CM point on $\mathcal{X}$. Therefore, there is an $(\mathcal{O}, \mu)$ Abelian surface with CM by $\mathcal{O}_K$, which we shall denote by (A, ι, λ). According to [5, Theorem 2.10], for any such (A, ι) there are K-CM elliptic curves E_1, E_2 such that $A \simeq E_1 \times E_2$. In particular it follows [Maybe I should add someting about K-CM points to the background] that

$$\mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q} \simeq M_{2\times 2}(K)$$

and hence from ι one can construct an embedding

$$\eta : \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Q} \simeq B \hookrightarrow M_{2\times 2}(K)$$

Therefore, according to the previous lemma, 3.1, $K \hookrightarrow B$, and hence according to [6, Proposition 14.6.7], K_2 and K_3 are fields, i.e. K does not split at 2 nor at 3 as claimed.

Assume that K does not split at 2 nor at 3, i.e. K_2 and K_3 are fields. According to [6, Proposition 14.6.7], K splits B. Therefore, due to [5, Proposition 2.12], there is a geometric $\mathcal{O}_K$-CM point on $\mathcal{X}$ as required. □

To draw the connection between this result and bielliptic Picard curves we will need the following corollary which follows from [4, Proposition 4.7]

Corollary 3.3. *Over $\mathbb{Q}\left(\sqrt{-3}\right)$ there is a 2 to 1 map*

$$\pi_2 : \mathcal{X} \to \mathcal{Y}$$

that maps a triple (A, λ, ι) to some triple of the form (A, λ, α) .

Using this map we can finally prove the promised theorem:

Theorem 3.4. *There exists a bielliptic Picard curve C with CM by $\mathcal{O}_K$ if and only if K does not split at 2 and does not split at 3 and $K \neq \mathbb{Q}[\sqrt{-3}]$.*

Proof. In [4, Remark 3.14], Laga and Shnidman show that the map

$$(C, \gamma) \mapsto \mathrm{Prym}(C, \gamma)$$

is an equivalence of categories

$$\left\{\begin{array}{c}\text{Marked bielliptic}\\ \text{Picard curves over } \overline{\mathbb{Q}}\end{array}\right\} \to \left\{\begin{array}{c}\text{Indecomposable}\\ \mu_3\text{-abelian surfaces over } \overline{\mathbb{Q}}\end{array}\right\}$$

Since $\mathcal{Y}$ is the coarse moduli space of μ_3-abelian surfaces over $\overline{\mathbb{Q}}$, we can compose π_2 restricted to the indecomposable points with a quasi inverse of the above categories equivalence functor, to obtain a map from $\mathcal{X}$ to the set of isomorphism classes of marked bielliptic Picard curves. However, by 3.2, there exists a geometric $\mathcal{O}_K$-CM point on $\mathcal{X}$ if and only if K doesn't split at 2 and doesn't split at 3. Therefore, it suffices to show that each of the maps in the above composition preserves the CM order and that the decomposable points on $\mathcal{Y}$ are exactly those with CM by $\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$. However, π_2 preserves the CM order because it doesn't change the surface itself and we have shown in 2.5 that the CM order depends only on the surface itself, and the inverse of the quasi inverse of the equivalence functor preserves the CM order because the CM order of a bielliptic Picard curve was defined to be the CM order of its Prym.

Finally, we will show that the decomposable points on $\mathcal{Y}$ are exactly those with CM by $\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$. According to the proof of [4, Lemma 4.3] for every decomposable μ_3 abelian surface (A, λ, α), A is isomorphic to the square of the elliptic curve $E : y^2 = x^3 + 1$. It is well known that the endomorphisms ring of E is $\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$ and hence $\mathrm{End}(A) = M_{2\times 2}(\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]})$ and hence $\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$ is embedded in the center of $\mathrm{End}(A)$ and hence it is contained in $\mathrm{End}_{\mathcal{O}}(A)$. However, $\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$ is by definition a maximal quadratic order, and hence, since as noted in the introduction $\mathrm{End}_{\mathcal{O}}(A)$ must be either $\mathbb{Z}$ or an order in an imaginary quadratic field, $\mathrm{End}_{\mathcal{O}}(A) = \mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$. All the points with CM by $\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$ are decomposable because according to [4, Remark 4.5] there are two decomposable points and according to the proof of [4, Lemma 6.17], there are two points of CM by $\mathcal{O}_{\mathbb{Q}[\sqrt{-3}]}$. □

Before moving to the next subject we would like to highlight another result that follows from the proof of the above theorem and will be useful in the following sections:

Corollary 3.5. *The equivalence of categories*

$$(C, \gamma) \mapsto \mathrm{Prym}(C, \gamma)$$

reduces to a map from CM marked bielliptic Picard curves to CM μ_3-abelian surfaces that preserves the CM order.

4. Upper Bound for the Field Generated by the j-invariant of a Bielliptic Picard Curve with CM

Keep the notation of π_2 from the previous section. Let (C, γ) be a marked bielliptic Picard curve. In [4, §2.7] Laga and Shnidman define the bigonal dual of (C, γ) which is another marked bielliptic Picard curve denoted by $(\hat{C}, \hat{\gamma})$ for which

$$\mathrm{Prym}(\widehat{C}) = \mathrm{Prym}(C)^{\vee}$$

(we will not provide here the details of how $\hat{\gamma}$ is defined). Accordingly they denote

$$\mathrm{Prym}\,(C, \gamma)^{\vee} := \mathrm{Prym}(\hat{C}, \hat{\gamma}) \tag{4.1}$$

Laga and Shnidman proved the following [4, Lemma 4.3]:

Lemma 4.1. *There exists a unique isomorphism $j : Y \to \mathbb{P}^1_{\mathbb{Q}}$ such that $j\left(\left[\mathrm{Prym}\,(C, \gamma)^{\vee}\right]\right) = j\,(C)$ for all marked bielliptic Picard curves (C, γ) over $\overline{\mathbb{Q}}$.*

It follows that:

Lemma 4.2. *For every marked bielliptic Picard curves (C, γ) over $\overline{\mathbb{Q}}$, $\mathbb{Q}\,(j\,(C)) = \mathbb{Q}\,(j\,[\mathrm{Prym}\,(C, \gamma)])$*

Proof. According to [4, Lemma 2.20],

$$j(\hat{C}) = \frac{1}{j(C)} \tag{4.2}$$

4.1 and 4.2 together imply that

$$j(C) = \frac{1}{j(\widehat{C})} = \frac{1}{j\left(\left[\mathrm{Prym}\left(\widehat{C},\hat{\gamma}\right)^{\vee}\right]\right)} = \frac{1}{j\left(\left[\mathrm{Prym}\left(\widehat{\widehat{C}},\hat{\hat{\gamma}}\right)\right]\right)} = \frac{1}{j\left(\left[\mathrm{Prym}\left(C,\gamma\right)\right]\right)}$$

and hence $j(C)$ and $j\left(\left[\mathrm{Prym}\left(C,\gamma\right)\right]\right)$ generate the same field over $\mathbb{Q}$. □

We thus reduced the question of determining the field generated by the j value of a bielliptic Picard curve to determining the field generated by the j value of a corresponding μ_3-abelian surface. Specifically, we may deduce the following

Theorem 4.3. *Let K be an imaginary quadratic field, let H be its Hilbert class field and let C be a bielliptic Picard curve with CM by $\mathcal{O}_K$, then*

$$K(j(C)) \subseteq H(\sqrt{-3})$$

Proof. According to corollaries 3.3 and 3.5 there is a point $P \in \mathcal{X}$ such that $\pi_2(P) \in \mathcal{Y}$ corresponds to $[\mathrm{Prym}(C)]$. Then according to 3.5, P also has CM by $\mathcal{O}_K$. Therefore from [5, Theorem 2.7] it follows that $K(P) = H$ and hence, according to 3.3, $\pi_2(P) \in H(\sqrt{-3})$. Thus, $j(\pi_2(P)) \in H(\sqrt{-3})$ and consequently, since by definition $K \subseteq H$, the last lemma, 4.2, implies that

$$K(j(C)) = K(j([\mathrm{Prym}(C))]) = K(j(\pi_2(P))) \subseteq H(\sqrt{-3})$$

as claimed. □

5. The Main Result

Recall that K is an imaginary quadratic field. We will begin with a lemma,

Lemma 5.1. *Let C/K be a smooth curve, let Φ be a Abelian group of automosphisms of C defined also over K, and let $\pi : C \to C/\Phi$ be the quotient map. For each $P \in C(\overline{K})$, define*

$$G_P = \left\{\sigma \in \mathrm{Gal}\left(\frac{K(P)}{K}\right) | \sigma(P) \in \Phi(P)\right\}$$

then G_P is a group and $K(\pi(P)) = K(P)^{G_P}$

Proof. 1 is clearly in G_P and given $\sigma, \tau \in G_P$ which correspond to $\phi, \psi \in \Phi$ respectively,

$$\sigma^{-1}(P) = \sigma^{-1}(\phi^{-1}(\phi(P))) = \sigma^{-1}(\phi^{-1}(\sigma(P))) = \sigma^{-1}(\sigma(\phi^{-1}(P))) = \phi^{-1}(P)$$
$$\sigma\tau(P) = \sigma(\psi(P)) = \psi(\sigma(P)) = \psi(\phi(P)) = \phi(\psi(P))$$

and in particular $\sigma^{-1}, \sigma\tau \in G_P$ and hence G_P is indeed a group. Let $\sigma \in G_P$, then

$$\sigma(\pi(P)) = \pi(\sigma(P)) = \pi(\phi(P)) = \pi(P)$$

and on the other hand, if $\eta \in \mathrm{Gal}(K[P]/K)$ is such that $\eta(\pi(P)) = \pi(P)$ then $\pi(\eta(P)) = \pi(P)$ and hence

$$\eta(P) \in \pi^{-1}(\pi(P)) = \Phi(P)$$

and hence $\eta \in G_P$. Therefore, $\mathrm{Gal}(K[x]/K[\pi(x)]) = G_P$. □

Let $n > 1$ be a square free integer such that $K = \mathbb{Q}[\sqrt{-n}]$, let $D = d_K$ and denote by H the Hilbert class field of K, and let $P \in \mathcal{X}$ be a point of complex multiplication by $\mathcal{O}_K$. For each $m \in \{2, 3\}$ denote by $\pi_m : \mathcal{X} \to \mathcal{X}/w_m$ the quotient map. Then using the last lemma we can prove the following corollary:

Corollary 5.2. *For each $m \in \{2, 3\}$ one of two cases occurs:*

- *$m \not| D$ and $w_m(P)$ is not conjugate to P over K.*
- *$m|D$ and $w_m(P) = \left(\frac{H/K}{\mathfrak{a}}\right)(P)$ where $\mathfrak{a}$ is the unique prime lying over m in K*

Proof. To avoid confusion with the field discriminant D, we will write D_A where Gonzalez and Rotger wrote D in [3]. According to their notation,

$$\begin{aligned} D_A =& 6 \\ N =& 1 \\ D_A(\mathcal{O}_K) =& \prod_{p|D_A, \left(\frac{\mathcal{O}_K}{p}\right)=-1} p = \frac{6}{\gcd(D, 6)} \\ N(\mathcal{O}_K) =& \prod_{p|N, \left(\frac{\mathcal{O}_K}{p}\right)=-1} p = 1 \end{aligned}$$

and hence

$$\frac{D_A N}{D_A(\mathcal{O}_K) N(\mathcal{O}_K)} = \frac{6}{\frac{6}{\gcd(D,6)} 1} = \gcd(D, 6)$$

and in particular, $m|\frac{D_A N}{D_A(\mathcal{O}_K)N(\mathcal{O}_K)}$ if and only if $m|D$. Therefore, according to [3, lemma 5.9], one of two cases occurs:

- $m \not| D$ and hence $P, w_m(P)$ are not conjugate over K.
- $m|D$ and hence there is an ideal $\mathfrak{a}$ in K whose norm is m and $w_m(P) = \left(\frac{H/K}{\mathfrak{a}}\right)$. However, by the second isomorphism theorem

  $$\frac{\mathbb{Z}}{\mathfrak{a} \cap \mathbb{Z}} \simeq \frac{\mathbb{Z} + \mathfrak{a}}{\mathfrak{a}} = \frac{\mathcal{O}_K}{\mathfrak{a}}$$

  and in particular

  $$\left|\frac{\mathbb{Z}}{\mathfrak{a} \cap \mathbb{Z}}\right| = \left|\frac{\mathcal{O}_K}{\mathfrak{a}}\right| = m$$

  and hence $\mathfrak{a}$ is the unique prime lying over m in $\mathbb{Z}$.

□

From this we can derive another corollary that will be useful soon. To state it, denote by $\mathfrak{a}, \mathfrak{b}$ the unique primes lying over $3, 2$ in K respectively. They are indeed unique because 3 and 2 don't split, so each of them is either inert or ramified and in any case there is indeed a unique prime lying over it. Denote by $\pi : \mathcal{X} \to \mathcal{X}^* := \mathcal{X}/W$ the quotient map, then:

Corollary 5.3.

$$\mathrm{Gal}\left(\frac{H}{K(\pi(P))}\right) = \left\langle \left(\frac{H/K}{\mathfrak{a}}\right), \left(\frac{H/K}{\mathfrak{b}}\right) \right\rangle$$

Proof. According to lemma 5.1,

$$K(\pi(P)) = K(P)^{G_P}$$

where G_P is defined as in lemma 5.1 and hence it suffices to show that

$$G_P = \left\langle \left(\frac{H/K}{\mathfrak{a}}\right), \left(\frac{H/K}{\mathfrak{b}}\right) \right\rangle \tag{5.1}$$

We will first show the inclusion from left to right. Let $0 \neq \sigma \in G_P$. Then one of two cases occurs:

- There is $m \in \{2,3\}$ such that $\sigma(P) = w_m(P)$ and hence, by corollary 5.2, $\simeq$ is the Artin symbol of the unique prime lying over m and in particular it belongs to the right hand side of equation 5.1.
- There is no such m. This means that $\sigma(P) = w_6(P)$. By the same argument as in the proof of corollary 5.2, it follows that $6|D$ and in particular $2,3|D$ and hence, by the content of corollary 5.2,

  $$\left(\frac{H/K}{\mathfrak{a}}\right), \left(\frac{H/K}{\mathfrak{b}}\right) \in G_P$$

  and hence

  $$\sigma(P) = w_6(P) = (w_2 \circ w_3)(P) = \left(\left(\frac{H/K}{\mathfrak{a}}\right) \circ \left(\frac{H/K}{\mathfrak{b}}\right)\right)(P) \tag{5.2}$$

  However, according to [5, Theorem 2.7], $K(P) = H$ and thus it follows from the last equation 5.2 that

  $$\sigma = \left(\left(\frac{H/K}{\mathfrak{a}}\right) \circ \left(\frac{H/K}{\mathfrak{b}}\right)\right) \in \left\langle \left(\frac{H/K}{\mathfrak{a}}\right), \left(\frac{H/K}{\mathfrak{b}}\right) \right\rangle$$

For the inclusion from right to left in equation 5.1 it's enough to show that for each $m \in \{2,3\}$ the Artin symbol of the unique prime lying over m in $\mathcal{O}_K$ is in G_P. Indeed, if $m \nmid D$ then m is not ramified in $\mathcal{O}_K$ and hence since it is not split either it is inert and in particular its Artin symbol is trivial and hence it trivilly belongs to G_P. On the other hand, if $m|D$ then by corollary 5.2 the image of P under its Artin symbol is $w_m(P)$ and hence its Artin symbol is indeed in G_P. □

According to [1, Theorem 6.1], for any odd prime $p|n$, $\mathbb{Q}[\sqrt{p^*}] \subseteq H$ when $p^* = (-1)^{\frac{p-1}{2}}$. Therefore, we can denote by

$$r_{p^*} : \mathrm{Gal}(H/K) \to \mathrm{Gal}(K[\sqrt{p^*}]/K)$$

the restriction homomorphism, by

$$\chi_{p^*} : \mathrm{Gal}\left(\frac{H}{K}\right) \to \{\pm 1\}$$

the unique such group homomorphism from $\mathrm{Gal}\left(\frac{H}{K}\right)$ to $\{\pm 1\}$ that factors through r_{p^*}. To analyze χ_{p^*} we will first prove the following lemma to describe $\mathcal{O}_{K[\sqrt{p^*}]}$ for any odd p.

Lemma 5.4. *Let p be an odd prime, then*

$$\mathcal{O}_{K[\sqrt{p^*}]} = \mathcal{O}_K\left[\frac{1+\sqrt{p^*}}{2}\right]$$

Proof. It's easy to see that, like in the classic quadratic case,

$$\mathcal{O}_{K[\sqrt{p^*}]} = \left\{\frac{\alpha+\beta\sqrt{p^*}}{2} | \alpha,\beta \in \mathcal{O}_K,\ \alpha \equiv \beta \bmod 2,\ \alpha^2 - p^*\beta^2 \equiv 0 \bmod 4\right\}$$

However, since $p^* \equiv 1 \bmod 4$, for any $\alpha,\beta \in \mathcal{O}_K$ such that $\alpha \equiv \beta \bmod 2$,

$$\alpha^2 = (2\gamma+\beta)^2 = 4\gamma^2 + 4\gamma\beta + \beta^2 \equiv \beta^2 \bmod 4$$

Therefore it follows that

$$\mathcal{O}_{K[\sqrt{p^*}]} = \left\{ \frac{\alpha + \beta\sqrt{p^*}}{2} | \alpha, \beta \in \mathcal{O}_K,\ \alpha \equiv \beta \bmod 2 \right\} = \mathcal{O}_K + \frac{1+\sqrt{p^*}}{2}\mathcal{O}_K$$

This group is obviously contained in $\mathcal{O}_K\left[\frac{1+\sqrt{p^*}}{2}\right]$ and the inclusion in the other direction is true because $\frac{1+\sqrt{p^*}}{2}$ and hence is an order over $\mathcal{O}_K$ and hence contained in the maximal such order which we already know to be $\mathcal{O}_K + \frac{1+\sqrt{p^*}}{2}\mathcal{O}_K$ □

Corollary 5.5. *Let $\mathfrak{a}$ be the unique prime lying over 2 in K and assume that it isn't inert (in K). Then for any odd prime p, $\mathfrak{a}$ is inert in $K[\sqrt{p^*}]$ if and only if $-1 = \left(\frac{2}{p}\right)$*

Proof. Since $\frac{1+\sqrt{p^*}}{2}$ generates $\mathcal{O}_{K[\sqrt{p^*}]}$ over $\mathcal{O}_K$, we can use Dedekind's factorization criterion to determine the factorization of $\mathfrak{a}\mathcal{O}_{K[\sqrt{p^*}]}$ based on the factorization of the image in $\frac{\mathcal{O}_K}{\mathfrak{a}}$ of the minimal polynomial of $\frac{1+\sqrt{p^*}}{2}$ over $\mathcal{O}_K$. The minimal polynomial is

$$f(x) = x^2 - x + \frac{1-p^*}{4}$$

since all its coefficients are in $\mathbb{Z}$, its image in $\frac{\mathcal{O}_K}{\mathfrak{a}}$ is its image in $\mathbb{F}_2$. Moreover, since $\mathfrak{a}$ is not inert, $\frac{\mathcal{O}_K}{\mathfrak{a}} \cong \mathbb{F}_2$, and the above polynomial is irreducible in $\mathbb{F}_2$ exactly when

$$0 \equiv 1 - p^* \bmod 8$$

which happens exactly when

$$p \equiv 1, 7 \bmod 8$$

and is separable otherwise. □

Equipped with this corollary we can prove another lemma. Let $N : K \to \mathbb{Q}$ be the norm function.

Lemma 5.6. *Let $\mathfrak{p}$ be prime in K, then for any odd $q|D$ such that $q \not| N(\mathfrak{p})$,*

$$\chi_{q^*}\left(\frac{H/K}{\mathfrak{p}}\right) = \left(\frac{N(\mathfrak{p})}{q}\right)$$

Proof. The lemma will follow by proving first that

$$\chi_{q^*} \circ \left(\frac{H/K}{-}\right) = \left(\frac{K[\sqrt{q^*}]/K}{-}\right) \tag{5.3}$$

and then that

$$\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)(\sqrt{q^*}) = \left(\frac{N(\mathfrak{p})}{q}\right)\sqrt{q^*} \tag{5.4}$$

for any $\mathfrak{p}$ and q as in the statement of the lemma.

To prove 5.3, let $\mathfrak{p} \subseteq \mathfrak{P} \subseteq \overline{\mathfrak{P}}$ be primes in $K \subseteq K[\sqrt{q^*}] \subseteq H$ respectively. It's enough to show that for any $\alpha \in \mathcal{O}_{K[\sqrt{q^*}]}$,

$$\left(\frac{H/K}{\mathfrak{p}}\right)(\alpha) - \alpha^{N(\mathfrak{p})} \in \mathfrak{P}$$

since this property defines $\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)$ uniquely. Indeed, for any $\alpha \in K[\sqrt{q^*}]$,

$$\left(\frac{H/K}{\mathfrak{p}}\right)(\alpha) - \alpha^{N(\mathfrak{p})} \in \overline{\mathfrak{P}}$$

by the definition of $\left(\frac{H/K}{\mathfrak{p}}\right)$, and $\left(\frac{H/K}{\mathfrak{p}}\right)(\alpha) \in K[\sqrt{q^*}]$ since $K[\sqrt{q^*}]$ is normal, and hence

$$\left(\frac{H/K}{\mathfrak{p}}\right)(\alpha) - \alpha^{N(\mathfrak{p})} \in \overline{\mathfrak{P}} \cap K[\sqrt{q^*}] = \mathfrak{P}$$

We shall now prove 5.4. Let again $\langle p \rangle \subseteq \mathfrak{p} \subseteq \mathfrak{P}$ be primes in $\mathbb{Q} \subseteq K \subseteq K[\sqrt{q^*}]$ respectively when $p > 0$, and let $q|D$ be an odd prime (in $\mathbb{Q}$) that doesn't divide $N(mfP)$. One of two cases occurs:

- p is inert in K and $N(\mathfrak{p}) = p^2$. In particular, $\mathfrak{p} = pK$ and hence $\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)$ is trivial and hence
$$\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)(\sqrt{q^*}) = \sqrt{q^*} = \left(\frac{p^2}{q}\right)\sqrt{q^*} = \left(\frac{N(\mathfrak{p})}{q}\right)\sqrt{q^*}$$
as claimed.
- p is either inert or ramified in K and $N(\mathfrak{p}) = p$. This is a bit trickier case. We will divide once again into cases:
  - Assume first that $2 \not| p$. Then
$$\begin{aligned}
&\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)(\sqrt{q^*}) \equiv \\
&(\sqrt{q^*})^p \equiv \\
&(q^*)^{\frac{p-1}{2}}\sqrt{q^*} \equiv \\
&(-1)^{\frac{q-1}{2}\frac{p-1}{2}} q^{\frac{p-1}{2}}\sqrt{q^*} \equiv \\
&(-1)^{\frac{q-1}{2}\frac{p-1}{2}}\left(\frac{q}{p}\right)\sqrt{q^*} \equiv \\
&\left(\frac{p}{q}\right)\sqrt{q^*} \equiv \\
&\left(\frac{N(\mathfrak{p})}{q}\right)\sqrt{q^*} \bmod \mathfrak{P}
\end{aligned}$$
and this congruence implies equality, since $2\sqrt{q^*} \notin \mathfrak{P}$ and hence $\sqrt{q^*} \not\equiv -\sqrt{q^*} \bmod \mathfrak{P}$.
  - Finally, assume that $p = 2$, the case for which we proved the previous lemma and corollary. It is enough to show that $\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)(\sqrt{q^*}) = \sqrt{q^*}$ if and only if $\left(\frac{2}{q}\right) = 1$. Indeed, $\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)(\sqrt{q^*}) = \sqrt{q^*}$ is equivalent to $\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right) = 1$. Therefore, it's enough to show that this condition is equivalent to $\left(\frac{2}{q}\right) = 1$. However, since $\mathrm{Gal}\left(\frac{\mathcal{O}_{K[\sqrt{q^*}]}/\mathfrak{P}}{\mathcal{O}_K/\mathfrak{p}}\right)$ is isomorphic to the decomposition group of $\mathbb{P}$, and $\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)$ corresponds under this isomorphism to a generator of $\mathrm{Gal}\left(\frac{\mathcal{O}_{K[\sqrt{q^*}]}/\mathfrak{P}}{\mathcal{O}_K/\mathfrak{p}}\right)$, $\left(\frac{K[\sqrt{q^*}]/K}{\mathfrak{p}}\right)$ is trivial if and only if $\mathrm{Gal}\left(\frac{\mathcal{O}_{K[\sqrt{q^*}]}/\mathfrak{P}}{\mathcal{O}_K/\mathfrak{p}}\right)$ is trivial i.e if and only if $\mathfrak{p}$ is not inert in $K[\sqrt{q^*}]$, and the last corollary, 5.5, shows that this happens exactly when $1 = \left(\frac{2}{q}\right)$ as claimed.

□

Corollary 5.7. *Let $p|D$ be an odd prime and let $\mathfrak{a}$ be an ideal in K such that $p \not| N(\mathfrak{a})$. Then*

$$\chi_{p^*}(\mathfrak{a}) = \left(\frac{N(\mathfrak{a})}{p}\right)$$

Lemma 5.8. *Let $\mathfrak{a}$ be the unique prime lying over p in K, then*

$$\chi_{p^*}\left(\frac{H/K}{\mathfrak{a}}\right) = \left(\frac{-D/p}{p}\right) = \left(\frac{n/p}{p}\right)$$

Proof. Let

$$\alpha = \frac{D + \sqrt{D}}{2}$$

Note that

$$N(\alpha) = \frac{D^2 - D}{4} = \frac{D(D-1)}{4}$$

and hence, since $\gcd(D, D-1) = 1$, $p \not| D-1$ and consequently $p \not| \frac{D(D-1)}{4p}$ and in particular

$$p \not| N\left(\frac{\langle\alpha\rangle}{\mathfrak{a}}\right)$$

Therefore,

$$1 = \chi_{p^*}(\langle\alpha\rangle) = \chi_{p^*}(\mathfrak{a})\chi_{p^*}\left(\frac{\langle\alpha\rangle}{\mathfrak{a}}\right)$$

and hence according to the last corollary

$$\begin{aligned}
&\chi_{p^*}(\mathfrak{a}) = \\
&\chi_{p^*}\left(\frac{\langle\alpha\rangle}{\mathfrak{a}}\right) = \\
&\left(\frac{N(\langle\alpha\rangle)/N(\mathfrak{a})}{p}\right) = \\
&\left(\frac{D/p}{p}\right)\left(\frac{(D-1)/4}{p}\right) = \\
&\left(\frac{D/p}{p}\right)\left(\frac{-1}{p}\right) = \\
&\left(\frac{-D/p}{p}\right)
\end{aligned}$$

as claimed. □

Before turning to prove the main result we need one last lemma

Lemma 5.9. *$H = H[\sqrt{-3}]$ if and only if $3|D$*

Proof. It's enough to prove that

$$K[\sqrt{-3}] \subseteq H \iff 3|D$$

The direction from right to left follows immediately from [1, Theorem 6.1]. Assume first that $K[\sqrt{-3}] \subseteq H$. Since 3 ramifies in $\mathbb{Q}[\sqrt{-3}]$, it does so also in $K[\sqrt{-3}]$. However, no prime lying above 3 in K ramifies in $K[\sqrt{-3}]$, and hence 3 also ramifies in K. Therefore $3|D$ as claimed. □

Recall from the introduction that we have defined

$$\phi : \mathrm{Gal}(H[\sqrt{-3}]/K) \to \mathrm{Gal}(K[\sqrt{-3}]/K) \times \mathrm{Gal}(H/K)$$

and denoted, under the identification between the groups $\mathrm{Gal}(K[\sqrt{-3}]/K)$ and $\{\pm 1\}$, the following elements of $\mathrm{Gal}(K[\sqrt{-3}]/K) \times \mathrm{Gal}(H/K)$:

$$\sigma = \begin{cases} \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right) & 2|D \\ 1 & 2 \not| D \end{cases}$$

$$\tau = \begin{cases} \left(1, \left(\frac{H/K}{\mathfrak{a}}\right)\right) & D \equiv 6 \bmod 9 \\ 1 & D \not\equiv 6 \bmod 9 \end{cases}$$

In the notation of the current context, we have promised to prove the following

Theorem 5.10. *Let C be a bielliptic Picar curve with CM by $\mathcal{O}_K$, and denote*

$$G = \mathrm{Gal}(H[\sqrt{-3}]/K[j(C)])$$

then

$$\phi(G) = \langle \sigma, \tau \rangle$$

However, according to lemma 4.2 and corollary 3.3 and since as discussed int he proof of theorem 3.4, the double cover $\pi_2 : \mathcal{X} \to \mathcal{Y}$ from corollary 3.3 preserves the CM order, it is enough to prove the following

Proposition 5.11. *Let $P \in \mathcal{X}$ be a point of CM by $\mathcal{O}_K$, and denote*

$$G = \mathrm{Gal}(H[\sqrt{-3}]/K[j(\pi_2(P))])$$

then

$$\phi(G) = \langle \sigma, \tau \rangle$$

Note that this proposition is technically even stronger since it does not ignore the points of CM by $\mathcal{O}_{\mathbb{Q}(\sqrt{-3})}$. let $P \in \mathcal{X}$ be a point of complex multiplication by $\mathcal{O}_K$ and as in the above proposition denote

$$G = \mathrm{Gal}(H[\sqrt{-3}]/K[j(\pi_2(P))])$$

In fact we will prove the following proposition which implies the one above due to the fact the D is never congruent to 1 modulo 3 since in this case 3 would be inert in K:

Proposition 5.12. *One of the following 6 cases occurs:*

- $D \equiv 5 \bmod 6$ *and* $\phi(G) = \{1\}$
- $D \equiv 2 \bmod 6$ *and* $\phi(G) = \left\langle \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right) \right\rangle$
- $D \equiv 15 \bmod 18$ *and* $\phi(G) = \left\langle \left(1, \left(\frac{H/K}{\mathfrak{a}}\right)\right) \right\rangle$
- $D \equiv 3 \bmod 18$ *and* $\phi(G) = \{1\}$
- $D \equiv 6 \bmod 18$ *and* $\phi(G) = \left\langle \left(1, \left(\frac{H/K}{\mathfrak{a}}\right)\right), \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right) \right\rangle$
- $D \equiv 12 \bmod 18$ *and* $\phi(G) = \left\langle \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right) \right\rangle$

We can make an even further reduction. According to lemma 5.9 in the last four cases $K[\sqrt{-3}] \subseteq H$ so the first coordinate which details the reduction of any automorphism in G to $K[\sqrt{-3}]$ is in fact redundant. In order to safely get rid of this coordinate, we need to make sure that in all of these four cases, the first coordinate corresponds to the reduction of the second coordinate to $K[\sqrt{-3}]$. Indeed, according to lemma 5.6, $\left(\frac{H/K}{\mathfrak{b}}\right)$ always reduces to -1 and according to 5.8, $\left(\frac{H/K}{\mathfrak{a}}\right)$ reduces to 1 exactly when $\frac{-D}{3} \equiv 1 \bmod 3$ which happens exactly in the two cases above in which $\left(\frac{H/K}{\mathfrak{a}}\right)$ appears. Therefore, we can, and will, prove the above proposition by proving the following one, where $j = j(\pi_2(P))$

Proposition 5.13. *One of the following 6 cases occurs:*

- $D \equiv 5 \bmod 6$ *and* $G = \{1\}$
- $D \equiv 2 \bmod 6$ *and* $\phi(G) = \left\langle \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right)\right\rangle$
- $D \equiv 15 \bmod 18$ *and* $\mathrm{Gal}(H/K[j]) = \left\langle \left(\frac{H/K}{\mathfrak{a}}\right)\right\rangle$
- $D \equiv 3 \bmod 18$ *and* $\mathrm{Gal}(H/K[j]) = \{1\}$
- $D \equiv 6 \bmod 18$ *and* $\mathrm{Gal}(H/K[j]) = \left\langle \left(\frac{H/K}{\mathfrak{a}}\right), \left(\frac{H/K}{\mathfrak{b}}\right)\right\rangle$
- $D \equiv 12 \bmod 18$ *and* $\mathrm{Gal}(H/K[j]) = \left\langle \left(\frac{H/K}{\mathfrak{b}}\right)\right\rangle$

Notice that we also slightly simplified the second case, as since ϕ is injective, the claim that $\phi(G)$ is trivial is equivalent to the claim that G itself is trivial. And finally we will present the multi-case proof.

Proof. To prove this proposition we will need to use more explicit models of $\mathcal{Y}$ and $\mathcal{X}$ and its different quotients. Following Elkies in [2, §3.1], let t be the unique rational function of degree 1 defined on $\mathcal{X}^*$ that maps the points of CM by the orders of discriminants $-3, -4, -24$ to $0, 1, \infty$ respectively. It follows that $\mathbb{Q}(\mathcal{X}^*) = \mathbb{Q}(t)$. Elkies shows in [2, §3.1] that

$$\mathbb{Q}(\mathcal{X}) = \mathbb{Q}(t)[\sqrt{-t}, \sqrt{3(t-1)}]$$

Additionally, Laga and Shnidman show in the proof of [4, Lemma 6.17] that

$$\mathbb{Q}(\mathcal{X}/w_3) = \mathbb{Q}(t)(\sqrt{-3t(t-1)})$$

so $\mathcal{X}/w_3$ has equation $s^2 = -3t(t-1)$. Moreover they showed that the involution $\overline{w_2}$ corresponds to the involution $(t, s) \mapsto (t, -s)$ i,e to the involution $(t, \sqrt{-3t(t-1)}) \mapsto (t, -\sqrt{-3t(t-1)})$. We will denote from now on

$$\begin{aligned} j =& j(P) \\ t =& t(P) \\ s =& s(P) \end{aligned}$$

Therefore, since is is shown in [4, Proposition 4.7] that $\mathcal{Y}$ is the twist of $\mathcal{X}/w_3$ along the involution $\overline{w_2}$ and extension $\mathbb{Q}(\sqrt{-3})$ and since j is an isomorphism $\mathcal{Y} \to \mathbb{P}^1$,

$$K(j) = K(t)\left(\frac{-\sqrt{-3t(t-1)}}{\sqrt{-3}}\right) = K(t)\left(\sqrt{t(t-1)}\right) \tag{5.5}$$

In particular it follows that

$$G \subseteq \mathrm{Gal}\left(\frac{H[\sqrt{-3}]}{K[t(P)]}\right)$$

Moreover, since

$$\sqrt{t(P)(t(P)-1)} = \frac{\sqrt{-3t(P)(t(P)-1)}}{\sqrt{-3}} = \frac{s(P)}{\sqrt{-3}}$$

if $s(P) \neq 0$ then G is the group of members $\sigma \in \mathrm{Gal}\left(\frac{H[\sqrt{-3}]}{K[t(P)]}\right)$ such that

$$\frac{\sqrt{-3t(P)(t(P)-1)}}{\sqrt{-3}} = \sigma\left(\frac{s(P)}{\sqrt{-3}}\right) = \frac{\sigma(s(P))}{\sigma(\sqrt{-3})}$$

i.e. such that σ either flips both $s(P)$ and $\sqrt{-3}$ or fixes both of them (Note that since $s(P) = \sqrt{-3t(P)(t(P)-1)}$ is quadratic over $K[t(P)]$, any $K[t(P)]$ automorphism of $\overline{\mathbb{Q}}$ either flips it or fixes it). In the two cases in which $0 = s(P)$, 5.5 shows that $K[j(P)] = K[t(P)]$ and hence G is trivial.

Additionally, note that

$$2|D \text{ and } \mathfrak{b} \text{ is principal} \implies D = -4 \tag{5.6}$$

Indeed, if $2|D$ then $\mathcal{O}_K = \mathbb{Z}[\sqrt{-n}]$ and 2 is ramified and hence if $\mathfrak{b}$ is principal, it means that there are $x, y \in \mathbb{Z}$ such that $\mathfrak{b} = \langle x + y\sqrt{-n}\rangle$ and hence

$$2 = |\mathfrak{b}| = N(x + y\sqrt{-n}) = x^2 + ny^2$$

when $N : K \to \mathbb{Q}$ is the norm map. Therefore, $ny^2 \leq 2$ and hence $n \leq \frac{2}{y^2} \leq 2$. However, if $n = 2$ then $\left(\frac{D}{3}\right) = \left(\frac{-8}{3}\right) = \left(\frac{1}{3}\right) = 1$ and consequently 3 splits in K which is a contradiction. The only option that is left is that $n = -1$ and consequently $D = -4$ as claimed.

We will start by a coarse division into cases based on whether 3 is ramified, which according to 5.9, determines whether $H[\sqrt{-3}] = H$. We will then subdivide each case into two sub-cases based on whether 2 is ramified which according to 5.3 will give as a better understanding of $\mathrm{Gal}\left(\frac{H}{K[t(P)]}\right)$ and some of the cases we will have to subdivide a bit further.

- Assume first that $3 \not| D$ and hence $H[\sqrt{-3}] \neq H$. In this case this actually completes the common analysis that the two subcases requires, so we will not postpone the sub division any further.
  - Assume that $2 \not| D$ or equivalently $D \equiv 5 \bmod 6$ and thus neither 2 nor 3 ramify in K. In particular $\left(\frac{H/K}{\mathfrak{a}}\right)$ and $\left(\frac{H/K}{\mathfrak{b}}\right)$ are both trivial and consequently, according to 5.3, $\mathrm{Gal}\left(\frac{H}{K[t(P)]}\right)$ is also trivial, or equivalently, $K[t(P)] = H$. Therefore, from the inclusion $G \subseteq \mathrm{Gal}\left(\frac{H[\sqrt{-3}]}{K[t(P)]}\right)$, shown above, it follows that $G \subseteq \mathrm{Gal}\left(\frac{H[\sqrt{-3}]}{H}\right)$. However, since $s(P) \in H$, this implies that any $\sigma \in G$ fixes $s(P)$ and hence must fix $\sqrt{-3}$ and thus fixes all of $H[\sqrt{-3}]$. Therefore G is trivial as claimed.
  - Assume that $2|D$ or equivalently $D \equiv 2 \bmod 6$ and thus 2 ramifies in K but 3 doesn't. If $\mathfrak{b}$ is principal then according to 5.6 $D = -4$ and one can check manually that this case satisfies the claim.

    Assume that $\mathfrak{b}$ is not principal. Therefore, $\left(\frac{H/K}{\mathfrak{b}}\right)$ is non-trivial. In particular, since $K[t(P)][s(P)] = H^{\left\langle\left(\frac{H/K}{\mathfrak{a}}\right)\right\rangle} = H$, any automorphism of H/K that fixes $s(P)$ is trivial, and hence $\left(\frac{H/K}{\mathfrak{b}}\right)$ flips $s(P)$.

Let now $\sigma \in G$ be non trivial. In particular, the reduction of σ to H is non trivial since it would imply that it fixes $s(P)$ and hence it must fix $\sqrt{-3}$ and hence it must be trivial in contradiction to the way we chose σ. Therefore $\sigma \restriction_H = \left(\frac{H/K}{\mathfrak{b}}\right)$. In particular, σ flips $s(P)$ and hence it also flips $\sqrt{-3}$ and hence $\phi(\sigma) = \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right)$. On the other hand, we know that there is $\sigma \in \mathrm{Gal}\left(\frac{H[\sqrt{-3}]}{K}\right)$ that satisfies this condition since $H \cap K[\sqrt{-3}] = K$ and hence ϕ is surjective, and this σ indeed fixes $k[t(P)]$ since so does $\left(\frac{H/K}{\mathfrak{b}}\right)$ which is its restriction to H. Therefore, $\phi(G) = \left\langle \left(-1, \left(\frac{H/K}{\mathfrak{b}}\right)\right)\right\rangle$ as claimed.

- Assume now that $3|D$ and hence $H[\sqrt{-3}] = H$. In particular, from the inclusion $G \subseteq \mathrm{Gal}\left(\frac{H[\sqrt{-3}]}{K[t(P)]}\right)$ it follows that

$$G \subseteq \mathrm{Gal}\left(\frac{H}{K[t(P)]}\right) = \left\langle \left(\frac{H/K}{\mathfrak{a}}\right), \left(\frac{H/K}{\mathfrak{b}}\right)\right\rangle \tag{5.7}$$

Furthermore, note that in this case, since $s(P) = \pi_3(P)$, according to 5.2, $\left(\frac{H/K}{\mathfrak{a}}\right)$ always fixes $s(P)$. Therefore, $\left(\frac{H/K}{\mathfrak{a}}\right) \in G$ exactly when it also fixes $\sqrt{-3}$ which according to 5.8 is exactly when $1 = \left(\frac{-D/3}{3}\right)$ or equivalently when $\frac{-D}{3} \equiv 1 \bmod 3$ or equivalently when $D \equiv 6 \bmod 9$. To sum it up,

$$\left(\frac{H/K}{\mathfrak{a}}\right) \in G \iff D \equiv 6 \bmod 9 \tag{5.8}$$

We will now divide into sub-cases:

- Assume first that $2 \nmid D$ or equivalently that 2 is inert. In particular $\mathfrak{b} = 2K$ and hence $\left(\frac{H/K}{\mathfrak{b}}\right)$ is trivial. Therefore, from 5.7 it follows that

$$G \subseteq \left\langle \left(\frac{H/K}{\mathfrak{a}}\right)\right\rangle$$

and hence it is only left to determine when $\left(\frac{H/K}{\mathfrak{a}}\right)$ is in G, and according to equation 5.8 one of two cases occurs:

  * $D \equiv 15 \bmod 18$ and $\left(\frac{H/K}{\mathfrak{a}}\right) \in G$ and hence $G = \left\langle \left(\frac{H/K}{\mathfrak{a}}\right)\right\rangle$ as claimed.
  * $D \equiv 3 \bmod 18$ and $\left(\frac{H/K}{\mathfrak{a}}\right) \notin G$ and hence $G = \{1\}$ as claimed.

- Assume finally that $2|D$. Assume first that $\mathfrak{a}$ and $\mathfrak{b}$ are equivalent. Therefore, there is $\alpha \in K$ such that $\mathfrak{a} = \alpha\mathfrak{b}$ and in particular

$$3 = \mathfrak{a}^2 = (\alpha\mathfrak{b})^2 = \alpha^2\mathfrak{b}^2 = 2\alpha^2$$

and in particular $\alpha^2 = u\frac{3}{2}$ when $u \in \mathcal{O}_K^*$. However, since $D < -4$ the only units in O_K are ± 1 and hence $\alpha^2 = \pm\frac{3}{2}$ or equivalently $(2\alpha)^2 = \pm 6$ and hence, since K is imaginary quadratic field $K = \mathbb{Q}[-6]$ and thus $D = -24$. In this case again the truth of the claim can be checked manually.

Assume from now on that $\mathfrak{a}$ and $\mathfrak{b}$ are not equivalent. Additionally since $3 \nmid -4$ and hence $D \neq -4$, according to 5.6, $\mathfrak{b}$ is not principal and in particular $\left(\frac{H/K}{b}\right)$ is not trivial. From the fact that $\left(\frac{H/K}{b}\right) \notin \left\{\left(\frac{H/K}{a}\right), 1\right\}$ it follows that $\left(\frac{H/K}{b}\right)$ flips $s(P)$.

Indeed, otherwise it fixes it and thus

$$H^{\left\langle\left(\frac{H/K}{a}\right)\right\rangle}=K[s(P)]\subseteq H^{\left\langle\left(\frac{H/K}{b}\right)\right\rangle}$$

and consequently

$$\left\langle\left(\frac{H/K}{b}\right)\right\rangle\subseteq\left\langle\left(\frac{H/K}{a}\right)\right\rangle$$

and hence since $\left(\frac{H/K}{a}\right)$ is of order at most 2, $\left(\frac{H/K}{b}\right)\notin\left\{\left(\frac{H/K}{a}\right),1\right\}$ so by contrapositive $\left(\frac{H/K}{b}\right)$ flips $s(P)$. However, according to 5.6, $\left(\frac{H/K}{b}\right)$ also flips $\sqrt{-3}$, and hence $\left(\frac{H/K}{b}\right)\in G$. Therefore, according to 5.8 one of two cases occurs:

* $D\equiv 6\mathrm{mod}\ 9$ or equivalently in this situation $D\equiv 6\mathrm{mod}\ 18$, and hence $\left(\frac{H/K}{a}\right)\in G$. Therefore, since we have shown that also $\left(\frac{H/K}{b}\right)\in G$,

$$G=\left\langle\left(\frac{H/K}{a}\right),\left(\frac{H/K}{b}\right)\right\rangle$$

as claimed.

* $D\equiv 3\mathrm{mod}\ 9$ or equivalently in this situation $D\equiv 12\mathrm{mod}\ 18$, and hence $\left(\frac{H/K}{a}\right)\notin G$. Therefore, since we have shown in contrast that $\left(\frac{H/K}{b}\right)\in G$,

$$G=\left\langle\left(\frac{H/K}{b}\right)\right\rangle$$

as claimed.

□

6. Some low degree j-invariants

In [4, Lemma 6.17] it is shown that

$$t=\frac{(j+1)^2}{4j} \tag{6.1}$$

The table below lists all the values of j that are mapped by the above formula to the values of t that are listed in [2, Table 1]. [2, Table 1] lists all the rational CM points on $\mathcal{X}^*$, and hence the table below lists all the values of j of points on $\mathcal{Y}$ whose t value is in $\mathbb{Q}$, all of them except ∞ also correspond to bielliptic Picard curves. Table [add table] also lists the discriminant of the CM order of each point, its t value and the field generated by its j-value (sorry for the quality of the table, I worked on it for hours and that is the best I could do).

D	t	j	K(j)
-3	$+\infty$	0	$Q[\sqrt{-3}]$
-4	1	1	$Q[\sqrt{-1}]$
-24	0	-1	$Q[\sqrt{-2\cdot 3}]$
-84	$-\frac{169}{27}$	$-\frac{1}{27}$	$Q[\sqrt{-3\cdot 7}]$
-40	$\frac{2312}{125}$	$\frac{1836}{125}\sqrt{6}+\frac{4499}{125}$	$Q[\sqrt{-2\cdot 5}\sqrt{-3\cdot 5}]$
-51	$-\frac{1377}{1024}$	$\frac{441}{512}\sqrt{17}-\frac{1889}{512}$	$Q[\sqrt{17}\sqrt{-3}]$
-19	$\frac{3211}{1024}$	$\frac{351}{512}\sqrt{57}+\frac{2699}{512}$	$Q[\sqrt{-19}\sqrt{-3}]$
-120	$\frac{5776}{3375}$	$\frac{125}{27}$	$Q[\sqrt{-2\cdot 3\cdot 5}]$
-52	$\frac{6877}{15625}$	$\frac{2484}{15625}\sqrt{-39}-\frac{1871}{15625}$	$Q[\sqrt{3}\sqrt{-13}]$
-132	$\frac{13689}{15625}$	$\frac{1}{2}i\sqrt{\frac{424030464}{244140625}}+\frac{11753}{15625}$	$Q[\sqrt{3\cdot 11}\sqrt{-1}]$
-168	$-\frac{701784}{15625}$	$\frac{579348}{15625}\sqrt{6}-\frac{1419193}{15625}$	$Q[\sqrt{2\cdot 3}\sqrt{-7}]$
-43	$\frac{212509987}{16000000}$	$\frac{930069}{8000000}\sqrt{129}+\frac{13250987}{8000000}$	$Q[\sqrt{-3}\sqrt{-43}]$
-228	$\frac{66863329}{11390625}$	$\frac{15625}{729}$	$Q[\sqrt{-3\cdot 19}]$
-88	$\frac{15545888}{20796875}$	$\frac{7377048}{20796875}\sqrt{-6}+\frac{10294901}{20796875}$	$Q[\sqrt{3\cdot 11}\sqrt{-2\cdot 11}]$
-123	$-\frac{296900721}{16000000}$	$\frac{47601099}{8000000}\sqrt{41}-\frac{304900721}{8000000}$	$Q[\sqrt{41}\sqrt{-3}]$
-312	$\frac{27008742384}{27680640625}$	$\frac{1364274072}{27680640625}\sqrt{-39}+\frac{26336844143}{27680640625}$	$Q[\sqrt{2}\sqrt{-3\cdot 13}]$
-67	$\frac{77903700667}{1024000000}$	$\frac{5458670217}{512000000}\sqrt{201}+\frac{77391700667}{512000000}$	$Q[\sqrt{-67}\sqrt{-3}]$
-148	$\frac{69630712957}{377149515625}$	$\frac{2777824249 2}{37714951 5625}\sqrt{-111}-\frac{237888089711}{377149515625}$	$Q[\sqrt{3}\sqrt{-37}]$
-372	$-\frac{455413074649}{747377296875}$	$-\frac{421875}{1771561}$	$Q[\sqrt{-3\cdot 31}]$
-408	$-\frac{3240860943673 6}{5596214 0625}$	$-\frac{4913}{11390625}$	$Q[\sqrt{-2\cdot 3\cdot 17}]$
-267	$-\frac{5766681714488721}{1814078464000000}$	$\frac{700848852120963}{907039232000000}\sqrt{89}-\frac{6673720946488721}{907039232000000}$	$Q[\sqrt{89}\sqrt{-3}]$
-232	$\frac{664322784834522 32}{5641323901282 8125}$	$\frac{21064782936601404}{5641323901282 8125}\sqrt{6}+\frac{76451317954076339}{5641323901282 8125}$	$Q[\sqrt{-2\cdot 29}\sqrt{-3\cdot 29}]$
-708	$\frac{7147575555484293036 9}{2243373273976038906 25}$	$\frac{1}{2}i\sqrt{\frac{1748143414841958940939440602885021506314 24}{5032723646389971721331457259013696289062 5}}-\frac{8138581628791802988 7}{2243373273976038906 25}$	$Q[\sqrt{3\cdot 59}\sqrt{-1}]$
-163	$\frac{6996902394513607050 67}{6841788140033440000 00}$	$\frac{4711121807269901751}{3420894070016720000 00}\sqrt{489}+\frac{3576008324496887050 67}{3420894070016720000 00}$	$Q[\sqrt{-3}\sqrt{-163}]$

7. Suggestions for further research

Lastly I include several suggestions for further research, maybe for myself:

- Calculate the field generated by j over $\mathbb{Q}$.
- Compare the values of our j-invariant with the classical j invariant.
- Generalize the results to other Shimura curves and quotients by different subgroups of the respective Atkin-Lehner groups.

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